Bruce Guo

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Summary

M.Sc. Computing Science student and Mitacs Fellow with a focus on Developer Productivity and Scalable AI Systems. Proven track record at Electronic Arts (EA) building high-performance tooling for large-scale Unreal Engine environments, including binary-to-textual conversion systems and distributed LLM agent architectures in Go. Specialized in bridging the gap between complex data structures (SQL/ElasticSearch) and actionable developer workflows.

Education

University of Alberta Master of Science (With Thesis)
Computing Science Edmonton, Canada • 2024/09/01 - Present

Research focused on the development of static analysis frameworks for Visual Programing Languages (VPLs). Engineered a custom methodology to adapt textual code metrics -- Line of Code (LOC), Halstead Volume, and Cyclomatic Complexity -- to node-based architectures (Unreal Engine Blueprint and Unity Visual Scripting).

Vanderbilt University Bachelor of Science
Computer Science Nashville, United States • 2021/09/01 - 2023/12/15

Experience

Electronic Arts (EA) Edmonton, Canada
Graduate Research Intern (Mitacs) 2024/05/01 - Present

- **Agentic RAG & LLM Orchestration:** Architected and implemented **distributed LLM agent strategies** in Go to automate complex report generation. Developed **Agentic RAG workflows** utilizing autonomous tool-calling to synthesize natural language prompts into actionable developer metrics.
- **Data Pipeline Engineering:** Architected an enrichment pipeline to ingest high-dimensional data into **Elasticsearch**; implemented a multi-stage lookup process to extract key identifiers, perform secondary queries for metadata retrieval, and merge disparate datasets into a unified, searchable index for real-time developer analytics.
- **High-Performance Tooling:** Developed a proprietary **binary-to-textual conversion engine** for Unreal Engine Blueprints, enabling high-speed HTML rendering of visual logic from SQL databases without requiring engine compilation.

University of Alberta Edmonton, Canada
Teaching Assistant 2024/09/01 - 2025/04/15

- **Technical Instruction:** Led weekly labs and office hours for **CMPUT 301 (Software Engineering)** and **CMPUT 404 (Web Development)**, assisting students with full-stack system design and debugging.
- **Code Assessment:** Evaluated weekly programming assignments, performing code reviews focused on design patterns, RESTful APIs, and version control (Git) best practices.
- **Student Support:** Provided hands-on troubleshooting for frontend frameworks, backend database integrations, and **robust object-oriented design patterns** in student-led software projects.

Project

ATex

- **Summary:** A high-performance, web-based LaTeX environment built to handle real-time document compilation and AI-assisted troubleshooting within a containerized microservices architecture.
- **Architecture:** Developed a responsive **React** and **FastAPI** web application using **Tectonic** for high-performance, containerized LaTeX compilation. Integrated **Supabase** for relational data storage and authentication.
- **System Design:** Engineered an asynchronous workflow using **Redis** to manage compilation task queues and integrated the **Gemini API** for AI-assisted error troubleshooting.